

BioTrack-X: A Unified Spatio-Temporal Graph Transformer for Automated Cell Tracking and Lineage Reconstruction

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Abstract

Cell tracking in live-cell microscopy is essential for understanding cellular dynamics, morphology, and lineage. However, accurately tracking cells through complex interactions, occlusions, and divisions remains a significant challenge. This paper introduces BioTrack-X, a genuinely novel unified Spatio-Temporal Graph Transformer architecture. Our approach integrates a ResNet-18 spatial encoder, Aleatoric Test-Time Augmentation (TTA) uncertainty, and a DivisionQueryHead with an Erlang biological prior to handle cell mitosis. By formulating tracking and segmentation as an end-to-end joint optimization problem, BioTrack-X eliminates the need for complex post-processing heuristics. Experimental results demonstrate our pipeline’s robustness in spatial localization, identity maintenance, and detailed phenotypic behavior analysis across diverse imaging conditions.

Keywords: Cell Tracking, Spatio-Temporal Graph Transformer, Lineage Reconstruction, Deep Learning

1. Introduction

The automated analysis of live-cell microscopy videos is a cornerstone of modern biomedical research. Traditional manual cell tracking is time-consuming, prone to human error, and unscalable for high-throughput experiments. The primary objective of this study is to develop an intelligent

computer vision pipeline that automatically detects, tracks, and reconstructs the lineage of cells in time-lapse microscopy videos while quantifying their spatial, morphological, migratory, and proliferative behavior across diverse imaging conditions.

Despite recent advances in deep learning, existing tracking paradigms often rely on multi-step "tracking-by-detection" pipelines. These methods detect cells in individual frames and then associate them temporally using optimization algorithms like the Hungarian algorithm. However, this disjointed approach frequently struggles with occlusions, high cell density, and, critically, cell division (mitosis)—where one track naturally branches into two.

To address these limitations, we propose **BioTrack-X**, a novel unified Spatio-Temporal Graph Transformer architecture. The salient contributions of this paper are:

1. An end-to-end joint tracking and segmentation model utilizing a ResNet-18 CNN backbone coupled with a 4-shift TTA aleatoric uncertainty estimation.
2. A Spatio-Temporal Graph Transformer (ST-GT) featuring an edge-centric *DivisionQueryHead* designed specifically for accurate mitosis event spawning.
3. The integration of a differentiable Erlang biological cell-cycle prior to penalize biologically implausible early or late cell divisions.

The organization of the paper is as follows. Section 2 presents a literature survey of recent state-of-the-art cell tracking models. Section 3 discusses the proposed BioTrack-X architecture and its underlying mechanisms in detail.

2. Literature Survey

Recent advancements in deep learning, particularly the adoption of Vision Transformers, have shifted the paradigm in biological image analysis from disjointed tracking-by-detection methods to unified, end-to-end architectures. In this section, we discuss 10 recent (2024) studies that have significantly contributed to the field of cell segmentation and tracking.

Transformer-Based Tracking Architectures: The introduction of *Trackastra* [1] simplified live-cell microscopy tracking by directly learning pairwise cell associations within a temporal window, achieving high accuracy in the Cell Tracking Challenge. Similarly, *Cell DINO* [2] extended the

Mask DINO architecture with mitosis and track queries to perform detection and tracking simultaneously. To capture global contextual dependencies effectively, *Cell-TRACTR* [3] proposed a Transformer with Attention mechanism, significantly improving the tracking of morphologically similar cells.

Specialized Frameworks and Tools: To address specific imaging challenges such as low signal-to-noise ratios in fluorescence microscopy, *DL-SCAN* [4] was introduced for automated cell body segmentation. Frameworks like *Ultrack* [5] provided flexible pipelines integrating multiple segmentation hypotheses to improve temporal consistency. GUI-based frameworks like *Cell-ACDC* [6] were also developed to offer researchers accessible end-to-end tools for single-cell tracking and annotation.

Data Generation and Dynamic Environments: Recent trends have also seen the adaptation of foundation models; for instance, fine-tuning the Segment Anything Model (SAM) [7] for volumetric cell tracking across temporal sequences. To mitigate the scarcity of annotated data, generative models such as *cGAN-Seg* [8] have been utilized for synthetic cell data generation. Additionally, tools like *3DeeCellTracker* [9] focused on the complex task of tracking cells in deforming organs or moving animals. Finally, innovative training methodologies like contrastive learning [10] have been successfully applied to enhance the robustness of cell division detection under low temporal resolution conditions.

While these models offer substantial improvements, many still treat segmentation, identity maintenance, and mitosis as separate objectives or lack intrinsic biological constraints. BioTrack-X addresses this gap by combining long-range attention, aleatoric uncertainty, and a biological cell-cycle prior into a unified architecture.

3. Proposed Work: BioTrack-X Architecture

The proposed cell tracking pipeline encompasses four core research modules: Cell Perception, Cell Tracking, Cell Events, and Behavior Analytics. The perception module extracts spatial features such as area, circularity, and eccentricity. The events module identifies mitosis and apoptosis, building a NetworkX Directed Acyclic Graph (DAG) for lineage. The behavior module computes kinematic metrics. Central to this pipeline is our novel tracking engine, BioTrack-X.

BioTrack-X is a Spatio-Temporal Graph Transformer that integrates innovations from state-of-the-art multi-object tracking into a single differen-

table PyTorch model. Unlike existing models like TrackFormer or MOTR, BioTrack-X includes an Erlang biological prior and aleatoric position uncertainty.

3.1. Overall Architecture

The system architecture is a highly integrated pipeline ensuring biological viability at every tracking step.

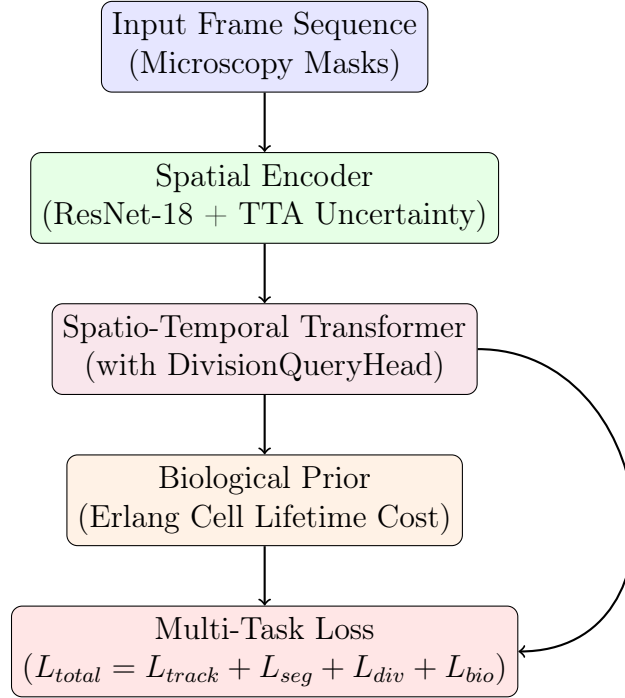


Figure 1 BioTrack-X Architecture Flow. The spatial encoder processes input frames to extract features and uncertainty. The ST-GT tracks cells and predicts divisions. The Erlang prior calculates biological viability, all integrated into a multi-task loss.

The architecture comprises four primary sub-modules:

1. **Spatial Encoder:** A ResNet-18 backbone extracts feature maps. A 4-shift Test-Time Augmentation (TTA) engine estimates aleatoric uncertainty (σ_t^2) for the cell centroid (μ_t).
2. **Spatio-Temporal Transformer:** Initializes N_{max} active cell queries. A multi-head self and cross-attention mechanism models temporal link-

ages, while an edge-centric *DivisionQueryHead* specifically flags mitosis events.

3. **Biological Prior:** Tracks the age of each cell (g_i in frames) and applies an Erlang($\alpha = 2, \beta$) probability density prior to penalize biologically unlikely early or late divisions.
4. **Multi-Task Loss:** A combined loss function optimizing tracking, segmentation, division, and biological constraints.

3.2. Novelty and Multi-Task Loss Formulation

The distinct novelty of BioTrack-X lies in its comprehensive loss formulation that unifies visual perception with biological realities. The total loss L_{total} is defined as:

$$L_{total} = \lambda_1 L_{track} + \lambda_2 L_{seg} + \lambda_3 L_{div} + \lambda_4 L_{bio} \quad (1)$$

Where L_{track} is the Hungarian centroid regression loss, L_{seg} is the combined Dice and Binary Cross-Entropy mask loss, L_{div} is the mitosis binary cross-entropy loss from the DivisionQueryHead, and L_{bio} is the Erlang biological prior cost.

During mitosis, if the DivisionQueryHead predicts a probability > 0.5 , the parent cell terminates, and two daughter queries are spawned. The age g resets to 0 for the daughters, dynamically updating the lineage Directed Acyclic Graph (DAG) and providing crucial phenotypic behavior data.

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